

Correlation and Simple Regression

AGRON 5130 - Unit 10 Assignment

Caio dos Santos

Introduction

The goal of this assignment is to help you practice the full workflow from Module 10:

- visualizing quantitative relationships
- interpreting correlation carefully
- fitting and interpreting a simple linear regression
- checking model assumptions
- making predictions and distinguishing interpolation from extrapolation

You will use a wheat–weed dataset collected in India. The key biological question is whether increasing weed pressure is associated with lower wheat yield.

Submit a knitted PDF/HTML generated from your R Markdown file to Canvas.

```
library(ggplot2)
```

```
weed <- read.csv("data/wheat_weed_india.csv")  
head(weed)
```

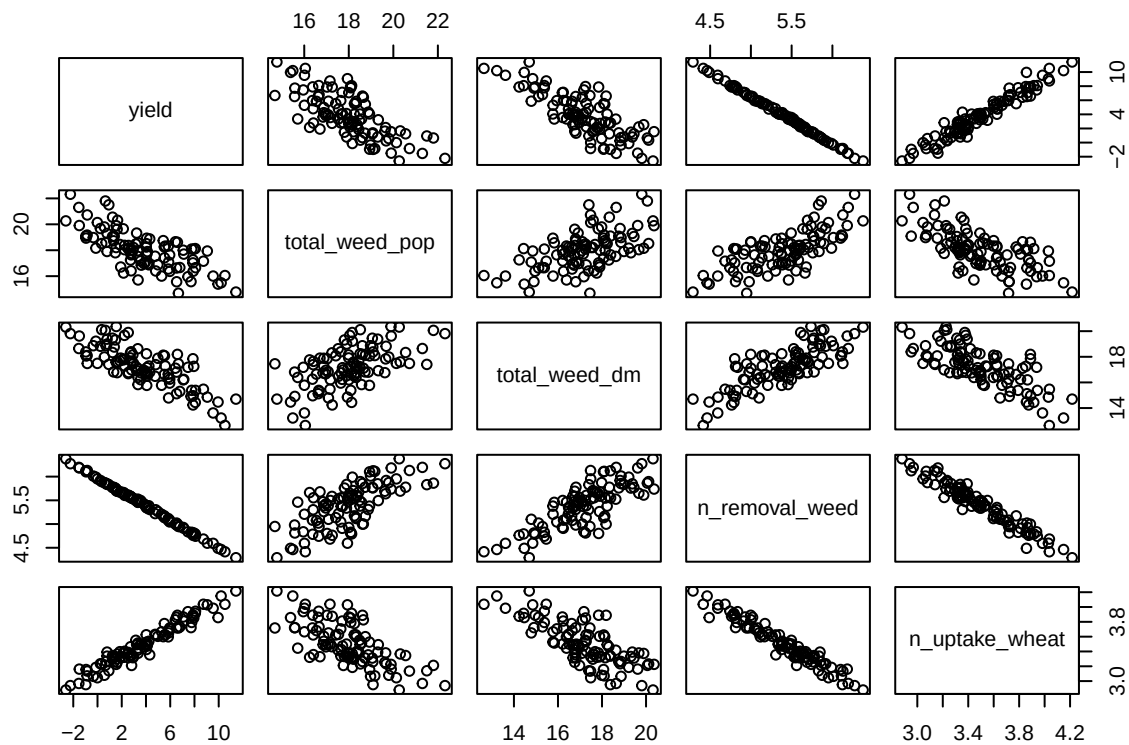
```
##      yield total_weed_pop total_weed_dm n_removal_weed n_uptake_wheat  
## 1 4.236967      16.87627      17.27988      5.385811      3.537604  
## 2 1.655525      20.31061      17.59948      5.694649      3.366266  
## 3 6.529892      15.56713      16.80601      4.979389      3.719664  
## 4 5.526334      17.41575      18.07812      5.135030      3.622897  
## 5 9.061101      17.94017      14.85238      4.596280      4.029803  
## 6 5.843537      16.54820      16.64260      5.076142      3.681508
```

Question 1

1 point

Create a scatterplot matrix for all quantitative variables in `weed`.

```
plot(weed)
```



Question 2

1 point

Create a correlation matrix for all quantitative variables in `weed`. Briefly identify which pair of variables shows the strongest association.

```
round(cor(weed), 2)
```

```
##           yield total_weed_pop total_weed_dm n_removal_weed n_uptake_wheat
## yield           1.00         -0.71         -0.79          -1.00           0.97
## total_weed_pop -0.71           1.00           0.57           0.71          -0.68
## total_weed_dm  -0.79           0.57           1.00           0.79          -0.75
## n_removal_weed -1.00           0.71           0.79           1.00          -0.97
## n_uptake_wheat  0.97          -0.68          -0.75          -0.97           1.00
```

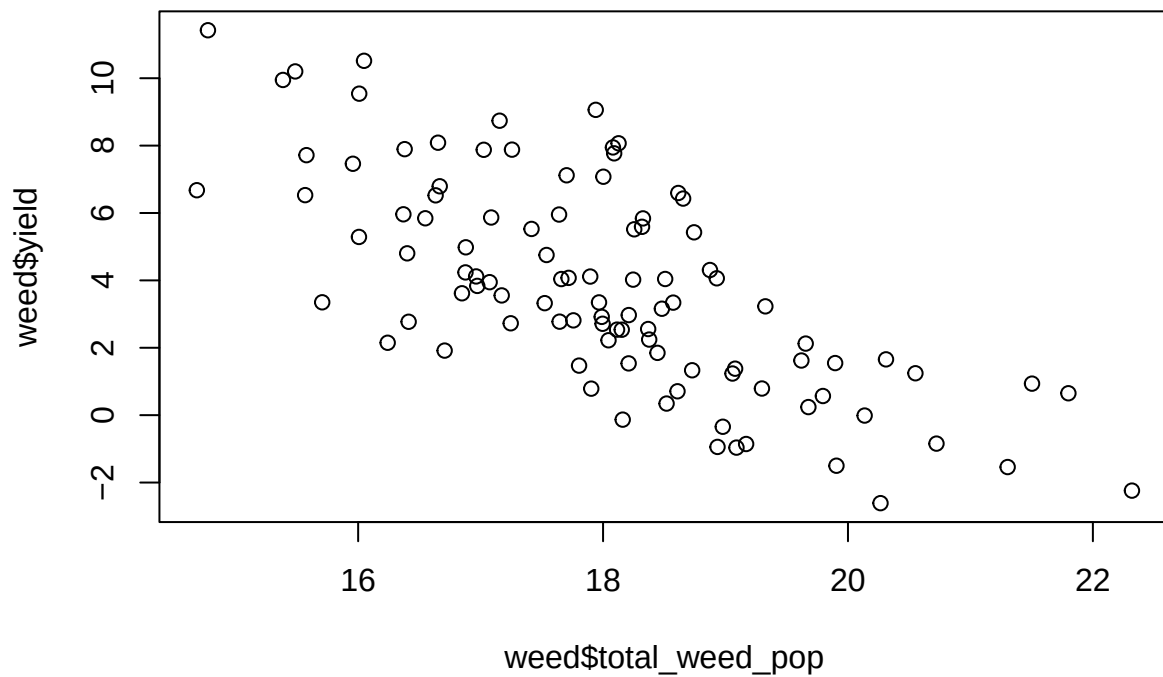
Question 3

1 point

Create a scatterplot showing the relationship between `total_weed_pop` and `yield`.

Optional but recommended: overlay a linear trend line.

```
plot(weed$total_weed_pop, weed$yield)
```



Question 4

1.5 points

Run a correlation test between `total_weed_pop` and `yield` using `cor.test()`.

```
cor.test(weed$total_weed_pop, weed$yield)
```

```
##
## Pearson's product-moment correlation
##
## data:  weed$total_weed_pop and weed$yield
## t = -10.079, df = 98, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.7980301 -0.6012840
## sample estimates:
##      cor
## -0.7134424
```

In 2–3 sentences, interpret:

1. direction of association
2. strength of association
3. whether this result implies causation

Question 5

1.5 points

Fit a simple linear regression model for `yield` as a function of `total_weed_pop`.

Display coefficient estimates using `coef()`.

```
model1 <- lm(yield ~ total_weed_pop, data = weed)
coef(model1)
```

```
##      (Intercept) total_weed_pop
##      30.429613      -1.477857
```

Question 6

1 point

Display R^2 for the model using `summary()`.

```
summary(model1)
```

```
##
## Call:
## lm(formula = yield ~ total_weed_pop, data = weed)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.2788 -1.3975 -0.3885  1.6223  5.1445
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    30.4296     2.6483   11.49  <2e-16 ***
## total_weed_pop  -1.4779     0.1466  -10.08  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.188 on 98 degrees of freedom
## Multiple R-squared:  0.509, Adjusted R-squared:  0.504
## F-statistic: 101.6 on 1 and 98 DF, p-value: < 2.2e-16
```

Question 7

2 points

Interpret the model from Questions 5–6.

Address all of the following:

1. Is the slope positive or negative?
2. Is the slope significantly different from zero?
3. What percentage of yield variation is explained by `total_weed_pop`?
4. In plain language, what does this model imply biologically?

Question 8

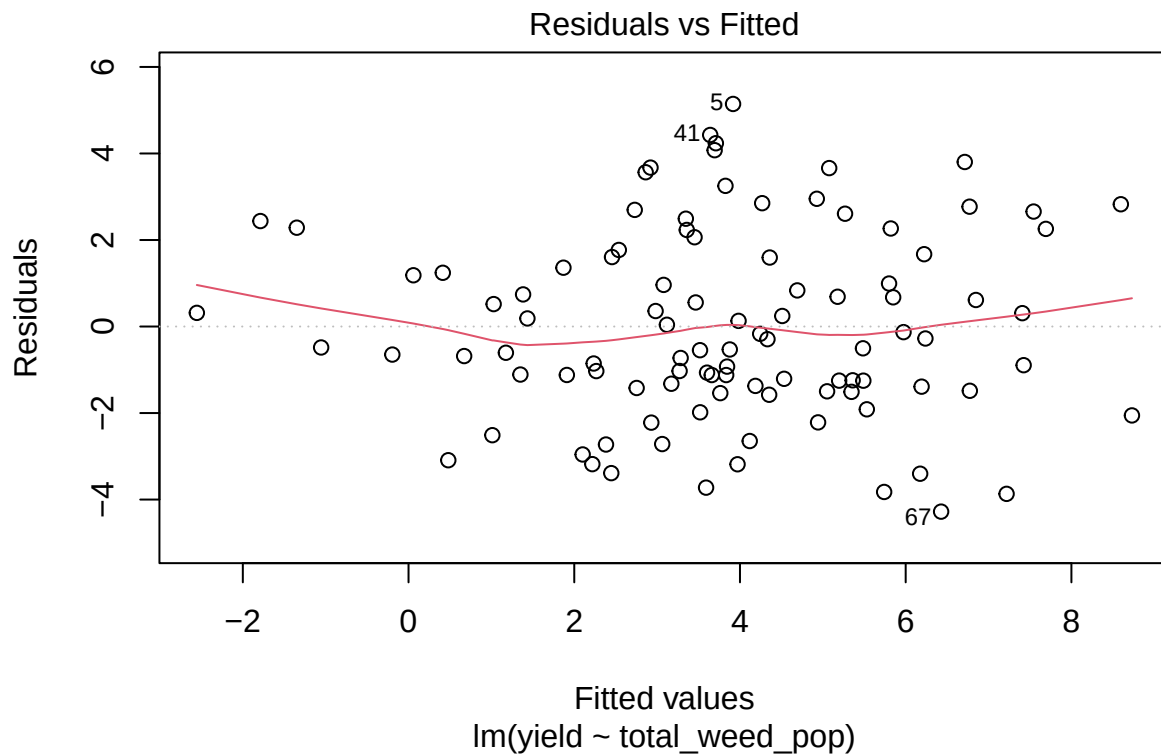
1.5 points

Check model assumptions for the regression from Question 5.

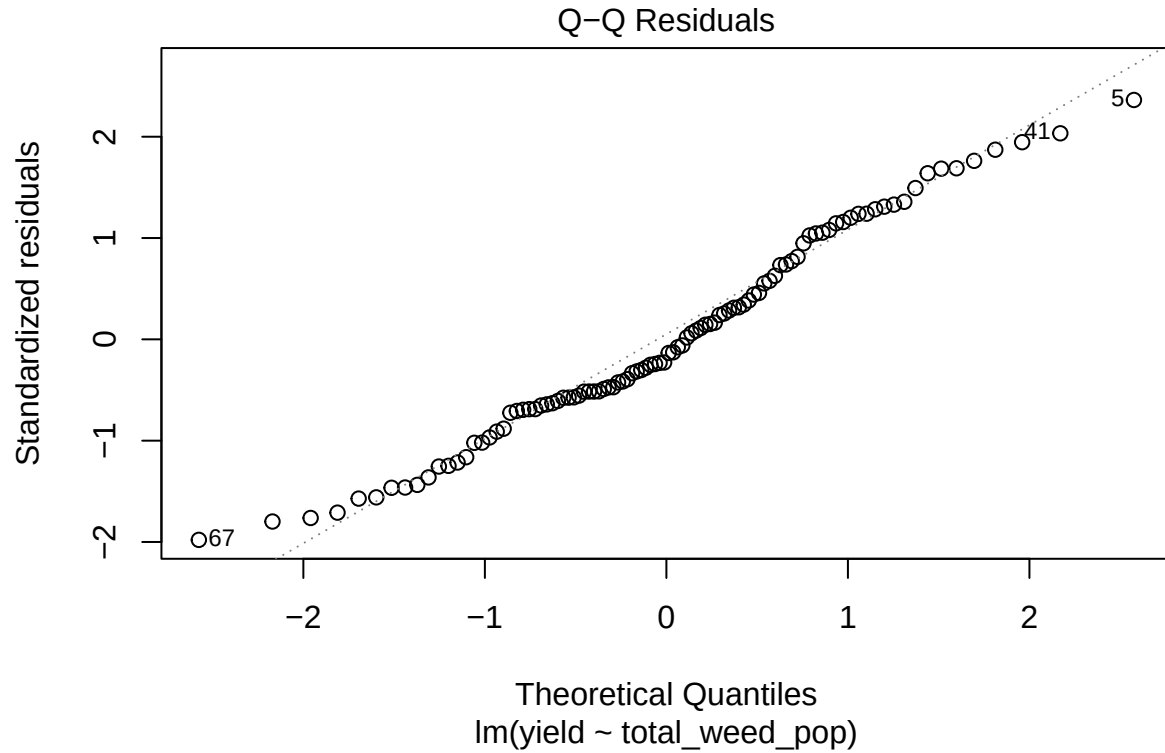
Produce:

1. residuals vs fitted plot (`which = 1`)
2. normal Q-Q plot (`which = 2`)

```
plot(model1, which = 1)
```



```
plot(model1, which = 2)
```



Then write 2-3 sentences on whether a straight-line model appears adequate.

Question 9

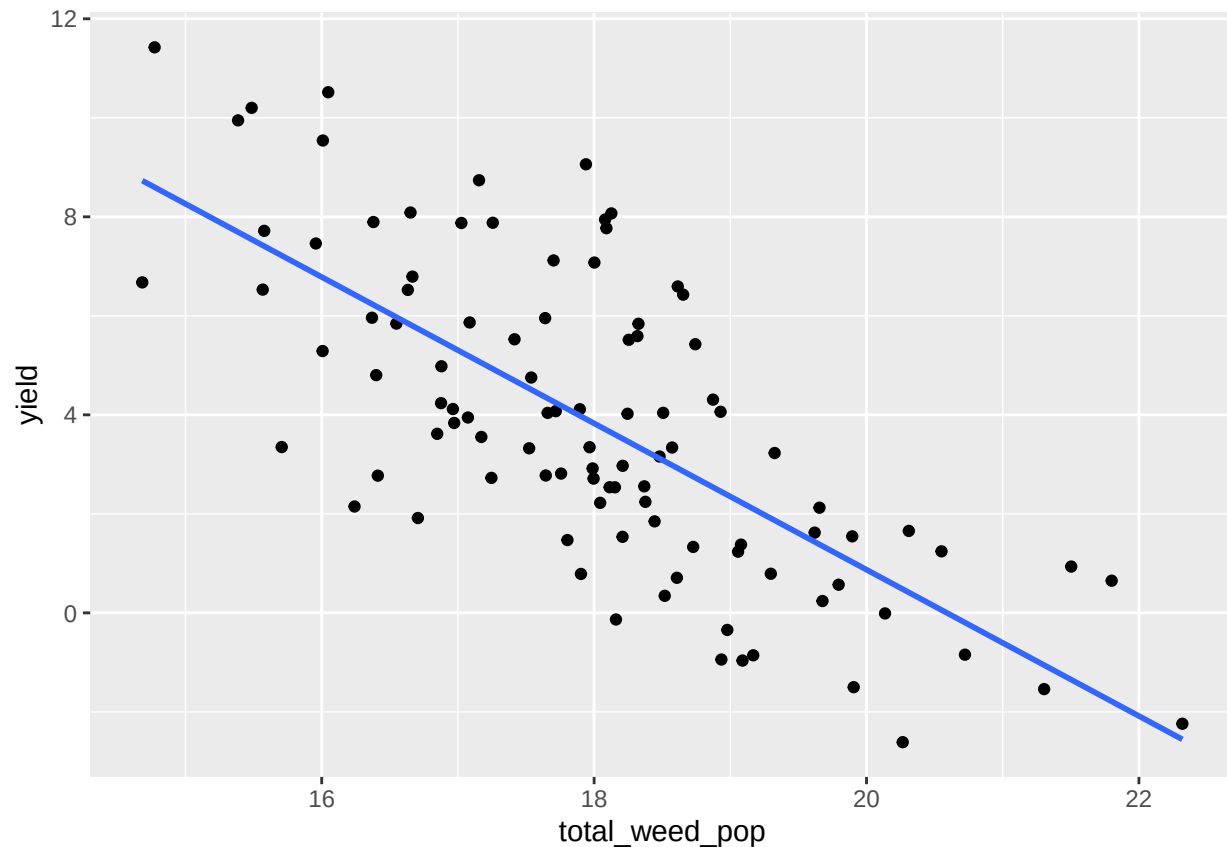
1 points

Using `ggplot()`, create a plot of `yield` vs `total_weed_pop` with:

1. observed points
2. fitted regression line

```
ggplot(weed, aes(x = total_weed_pop, y = yield))+  
  geom_point()+  
  geom_smooth(method = 'lm', se = F)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Question 10

2 points

Using the `new_weed_pops` data frame below, predict yield for each value of `total_weed_pop`.

Note: negative values are included intentionally to explore extrapolation.

```
new_weed_pops <- data.frame(
  total_weed_pop = c(-1, 1, 3, 5, 7)
)
```

Generate:

1. predicted values
2. 95% confidence intervals for the mean response

Hint: use `predict(..., interval = "confidence")`

```
predict(model1, newdata = new_weed_pops, interval = 'conf')
```

```
##      fit      lwr      upr
## 1 31.90747 26.36205 37.45289
## 2 28.95176 23.98627 33.91724
```

```
## 3 25.99604 21.60997 30.38211
## 4 23.04033 19.23290 26.84776
## 5 20.08461 16.85464 23.31459
```

Question 11

1.5 points

From Question 10, classify each prediction as:

1. interpolation
2. extrapolation

Justify your classification using the observed range of `total_weed_pop` in the original data.

Which predictions would you trust the most? Why?

Optional Bonus Question

+1 point

Based on your residual plots, do you see any signs that the assumptions of the linear model may be violated?

If so, what alternative approach might be more appropriate?